Analysis G9: testing a biomarker-by-drug interaction in an aggregated N-of-1 trial with carryover

pmsimstats team

Purpose. G9 is the analysis we recommend for the biomarker-by-drug interaction in a Hybrid aggregated N-of-1 trial when carryover is present and the carryover half-life is unknown. It was the best of the analyses evaluated in a staged simulation under the correlation encoding of the interaction.

Model. For participant i at visit t, with w_{it} the week of the visit,

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_w w_{it} + \beta_D D_{bc,it}(h) + \beta_{BD}\, B_i\, D_{bc,it}(h) + u_i + \varepsilon_{it},$$

$$u_i \sim N(0, \sigma_u^2), \qquad \mathrm{Corr}(\varepsilon_{it}, \varepsilon_{is}) = \varphi^{|w_{it} - w_{is}|}, \qquad D_{bc,it}(h) = \begin{cases} 1 & \text{on drug} \\ 2^{-t_{sd,it}/h} & \text{off drug after exposure} \\ 0 & \text{before first exposure.} \end{cases}$$

The half-life is chosen from the data, $\hat{h} = \arg\min_{h \in \mathcal{T}} \mathrm{AIC}\{M(h)\}$ with $\mathcal{T} = \{0.25, 0.5, 1, 2\}$ weeks and AIC from maximum-likelihood fits. The selected model is refitted by restricted maximum likelihood. The interaction is tested with $T = \hat{\beta}_{BD} / \mathrm{SE}_{CR2}(\hat{\beta}_{BD})$ against a t distribution with Satterthwaite degrees of freedom, where

$$\mathrm{SE}^2_{CR2} = \Big[(X^\top W X)^{-1} \sum_i X_i^\top W A_i\, \hat{e}_i \hat{e}_i^\top A_i^\top W X_i\, (X^\top W X)^{-1} \Big]_{BD,BD}, \qquad A_i = (I - H_{ii})^{-1/2}.$$

Notation. B_i: baseline biomarker, centered. $D_{bc,it}$: exposure-decayed drug indicator. $t_{sd,it}$: weeks since the last discontinuation (0 before exposure). h: analysis half-life; $\hat{h}$: its selected value. u_i: participant random intercept. ε_{it}: residual with continuous-time AR(1) correlation φ. β_{BD}: the interaction, the change in drug effect per unit of biomarker (under the correlation encoding $-c_{bm}\sigma_{BR}$ per standard deviation of the biomarker at the true half-life). X_i, W, $\hat{e}_i$, H_{ii}: participant design matrix, working weights, residuals, and leverage block. CR2: the bias-reduced cluster-robust variance estimator (Bell and McCaffrey).

Why G9. The ladder adds one change at a time (original analysis E0; G1 binary drug with AR(1); G6 adds CR2; G8 adds exposure decay at the true half-life; G9 selects the half-life). Pooled over the corrected original grid (Hybrid, OL+BDC and CO; N of 35 and 70):

Scenarios tested (G9 in every one; each with a null and a power cell unless noted). Designs: Hybrid, OL+BDC and crossover, N of 35 and 70, half-lives 0, 0.5 and 1 week. Misspecified decay: Weibull shapes 0.5 and 2 analyzed as exponential; true half-lives 0.2, 2 and 4 weeks. Serial correlation 0.5 and 0.9. Dropout: completely at random and dependent on baseline severity, 10, 20 and 30 percent. Prognostic biomarkers: correlation with the placebo response, the natural course and baseline severity, 0.3 and a feasible ladder to 0.8. Large N of 140 and 280 (null only). A biomarker that predicts the natural-history trend and a patient-specific slope. A Hybrid run with eight null and eight power cells at 5,000 and 1,000 replicates.

- **Size.** G9 rejects at 0.056 under the null, against 0.110 for E0, which is liberal, and 0.038 for G1, which is conservative. Across seven scenario families (misspecified decay, serial correlation, dropout, prognostic pathways, large N, trend) its pooled size is 0.045 to 0.060.
- **Power.** G9 has 0.641 power, the same as G8 (0.640), which needs the true half-life. It exceeds G1 (0.563) and F1 (0.517). E0 has higher raw power only because it rejects too often.
- **No pharmacokinetic input.** The analyst supplies a grid, not a half-life. Power matches the true-half-life analysis even though the selected value is spread over the grid.
- **Costs, stated.** Selection adds 0.0059 (paired SE 0.0013) to size over G8 (0.055 against 0.049 pooled over the null cells), and its 95% interval covers about 0.93 to 0.94 of the time. A fixed 1-week half-life with CR2 avoids selection, holds nominal size, and stays within two points of G8's power.
- **Limits.** The results hold under the correlation encoding, in the Hybrid design, on one parameter set. A biomarker that predicts the natural-history trend inflates the size of every analysis examined; G9 rejects at 0.074 at $N = 70$.

Analysis	What changes	Null rejection (MCSE)	Power at $c_{bm} = 0.45$, $t_{1/2} = 1$ (MCSE)
E0	Original analysis	0.110 (0.006)	0.691 (0.008)
G1	Binary drug, AR(1)	0.038 (0.003)	0.563 (0.009)
G6	+ CR2 inference	0.055 (0.004)	0.587 (0.009)
G8	+ exposure decay, true h	0.050 (0.004)	0.640 (0.009)
G9	+ selected $\hat{h}$ (recommended)	0.056 (0.004)	0.641 (0.009)
F1	Expectancy terms, fixed h	0.053 (0.005)	0.517 (0.011)